Quantitative Aptitude — Company-style Practice Set

Time & Work | Time, Speed & Distance | Profit & Loss

Q1. Time & Work

Category: Amazon | **Difficulty:** Medium

Question: A can complete a job in 10 days and B can complete the same job in 15 days. They start working together. After 3 days, A leaves the work. How many more days will B take to complete the remaining work?

Solution:

A's 1-day work = 1/10, B's 1-day work = 1/15.

Combined 1-day work = 1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6.

Work done together in 3 days = 3 × 1/6 = 1/2.

Remaining work = 1 − 1/2 = 1/2.

B alone completes 1/15 of the work per day.

Time for B to finish remaining 1/2 work = (1/2) ÷ (1/15) = 15/2 = 7.5 days.

Answer: 7.5 days

Q2. Time & Work

Category: TCS Digital | **Difficulty:** Medium

Question: A and B together can complete a piece of work in 6 days. A alone can complete the same work in 10 days. How long will B alone take to complete the work?

Solution:

(A+B)'s 1-day work = 1/6.

A's 1-day work = 1/10.

B's 1-day work = 1/6 − 1/10 = 5/30 − 3/30 = 2/30 = 1/15.

So B alone takes 15 days to complete the work.

Answer: 15 days

Q3. Time, Speed & Distance

Category: Google | **Difficulty:** Medium

Question: A train of length 120 meters crosses a pole in 12 seconds. What is the speed of the train in km/h?

Solution:

Distance covered while crossing a pole = length of train = 120 m.

Time taken = 12 seconds.

Speed = Distance / Time = 120 / 12 = 10 m/s.

Convert m/s to km/h: multiply by 18/5.

Speed = 10 × 18/5 = 36 km/h.

Answer: 36 km/h

Q4. Time, Speed & Distance

Category: Uber | **Difficulty:** Medium

Question: A man travels 30 km in 2 hours and then covers another 40 km in 3 hours. What is his average speed for the entire journey?

Solution:

Total distance = 30 + 40 = 70 km.

Total time = 2 + 3 = 5 hours.

Average speed = Total distance / Total time = 70 / 5 = 14 km/h.

Note: Average speed is total distance ÷ total time, not the simple average of the two individual speeds.

Answer: 14 km/h

Q5. Profit & Loss

Category: Amazon | **Difficulty:** Medium

Question: A shopkeeper sells an article at 20% profit. If he had bought it at 10% less and sold it for Rs 20 less, he would still have made 20% profit. What was the original cost price?

Solution:

Let original cost price = Rs x.

Original selling price (20% profit) = 1.2x.

New cost price (10% less) = 0.9x.

New selling price (still 20% profit on new CP) = $1.2 \times 0.9x = 1.08x$.

Also, new selling price = original SP − 20 = 1.2x − 20.

Equating: $1.08x = 1.2x - 20 \Rightarrow 0.12x = 20 \Rightarrow x = 20 / 0.12 = 166.67$.

Check: CP = 166.67, SP = 200 (20% profit). New CP = 150, new SP = $1.2 \times 150 = 180 = 200 - 20$. ✓

Answer: Rs 166.67 (≈ Rs 500/3)